Customer Support Agent Evaluation Report

Evaluation journey findings prompt improvement trace analysis and LLM judge calibration

150	94.0%	99.3%	9 of 9
Labeled tickets	Baseline accuracy	Improved accuracy	Target failures fixed

Project Goal and Context

This project evaluates an e-commerce customer-support routing agent that assigns each incoming ticket to one of five operational queues: order_status, refund_request, product_issue, account_help, or other. Accurate routing matters because a wrong classification sends the customer to the wrong team, creating avoidable transfers, slower resolution, and additional support effort.

Our goal was not only to improve this classifier, but to learn and demonstrate a repeatable agent-evaluation process that can be applied to future AI systems. Using a fixed 150-ticket labeled dataset, we reviewed the ground truth, established a reproducible baseline, annotated and clustered recurring failures, selected one high-impact category, made one focused prompt change, and reran the same cases to measure improvements and regressions. LangSmith traces connected aggregate metrics to individual decisions, while an LLM judge was calibrated against human labels. This report documents both the measured result and the evaluation workflow that produced it.

Model	Nebius openai gpt oss 120b
LangSmith project	customer support evals
Evaluation date	September 6 2026

Executive Summary

The controlled prompt change improved routing accuracy from 94.0 percent to 99.3 percent. It corrected all nine Account Workflow Missed failures and introduced one regression. The improved run met the declared quality and latency thresholds, although average token usage increased by 24.4 percent.

Evaluation One Liner

We measured exact match routing accuracy, account workflow failure count, regression count, latency, and token usage on a fixed 150 ticket labeled dataset. Human reviewed labels supplied the ground truth, and LangSmith connected the baseline and improved results to individual model traces.

Evaluation Flow

Stage	Work completed	Output
1	Reviewed the supplied labels and isolated 29 reference failures	Trusted evaluation scope
2	Annotated each failure and grouped recurring patterns	Three failure categories
3	Ran the unchanged prompt on the same 150 tickets	94.0 percent baseline
4	Added one account workflow disambiguation rule	Controlled prompt version 2
5	Reran the same cases and checked every changed prediction	99.3 percent and one regression
6	Compared two LLM judges against human labels	100 percent agreement for both

Main Finding

The original account help definition named login, password, address, and payment method changes but omitted account linked workflow failures. The new rule gave the model an explicit policy for app crashes, rejected checkout promo codes, repeated logout, and purchases missing from account history.

Evaluation Design

Agent and User Outcome

The classifier predicts one of five queues: order_status, refund_request, product_issue, account_help, or other. The user needs the ticket to reach the team that can resolve the primary problem without an avoidable transfer or delay.

Fixed Dataset

Category	Cases	Share	Coverage
order_status	24	16%	Tracking delivery status and non delivery
refund_request	42	28%	Refunds subscriptions and price adjustments
product_issue	24	16%	Damaged wrong defective or not as described
account_help	30	20%	Login checkout app access and order history
other	30	20%	General questions feedback and browsing
Total	150	100%	Direct ambiguous and known failure cases

Ground Truth Review

We treated the supplied labels as the assignment rubric and reviewed them during failure annotation. Some refund cases require subjective tie breaking between the underlying problem and the requested remedy. We retained those cases but recognized them as higher noise. The account workflow labels were more internally consistent.

Metrics and Pass Bars

Metric	Method	Pass bar
Overall routing accuracy	Exact match	At least 95%
Account workflow failures	Exact match and human grouping	0 after improvement
Regressions	Per ticket comparison	No more than 1
Latency	LangSmith duration	p95 below 1.5 seconds
Token usage	LangSmith totals	Report as cost proxy

Trace Design

LangSmith recorded one parent trace per ticket and child spans for prompt construction, the Nebius model call, the runnable sequence, and structured output parsing. Ticket ID, run name, prompt version, true category, prediction, and reasoning connect aggregate metrics to individual cases. Correctness is stored in the paired result row.

Reference Failure Analysis

The instructor workbook recorded 121 correct predictions out of 150, or 80.7 percent accuracy, with 29 failures. We used this run for qualitative failure discovery. We did not use it as the numerical baseline for the prompt comparison because its model configuration differed from our Nebius run.

Failure category	Count	Pattern	Business consequence
Refund Intent Missed	12	Money back intent loses to delivery policy or account cues	Financial requests reach the wrong team
Account Workflow Missed	9	Checkout app access promo code or account history is routed by surface topic	Customers can be blocked from paying or accessing purchases
Scope Boundary Misread	8	A keyword overrides the actual operational scope	Specialist queues receive avoidable transfers

Improvement Target

Refund Intent Missed was the largest cluster, but several examples depended on inconsistent tie breaking. Account Workflow Missed contained nine consistently labeled high impact cases, so it provided the cleaner controlled test.

Reproducible Baseline

Measure	Result
Correct predictions	141 of 150
Accuracy	94.0%
p50 latency	0.603 seconds
p95 latency	0.936 seconds
Total tokens	45,145
Average tokens per ticket	301.0

All nine baseline failures were true account_help tickets. Three involved app or order history access, three involved rejected promo codes, and three involved purchases missing from account history.

Prompt Improvement and Results

Improvement Hypothesis

An explicit account workflow rule should move the nine target cases to account_help without broadly over routing unrelated order product refund or general inquiry cases.

Prompt Change

> *If the customer cannot complete an account linked action because of repeated logout, an app or site crash, a rejected promo code at checkout, or a missing order record in account history, classify as account_help. Do not route only from surface words such as order, product, or discount.*

The dataset, model, temperature, structured output schema, evaluator logic, and trace structure remained unchanged. This rule was the only substantive classifier change.

Measured Result

Measure	Baseline	Improved	Delta
Correct predictions	141 of 150	149 of 150	+8
Accuracy	94.0%	99.3%	+5.3 points
Target failures	9	0	-9
Regressions	0	1	+1
p50 latency	0.603 s	0.574 s	-0.029 s
p95 latency	0.936 s	0.997 s	+0.061 s
Total tokens	45,145	56,161	+11,016
Average tokens	301.0	374.4	+73.4

Interpretation

The revised prompt met the quality regression and latency pass bars. It fixed all nine targeted failures and increased accuracy by 5.3 percentage points. Average token usage increased by 24.4 percent, which is the main operational tradeoff.

LangSmith Baseline Trace

Ticket t105 was an account help case about an app crash while viewing order history. The baseline predicted other because the original category definition did not cover the workflow.

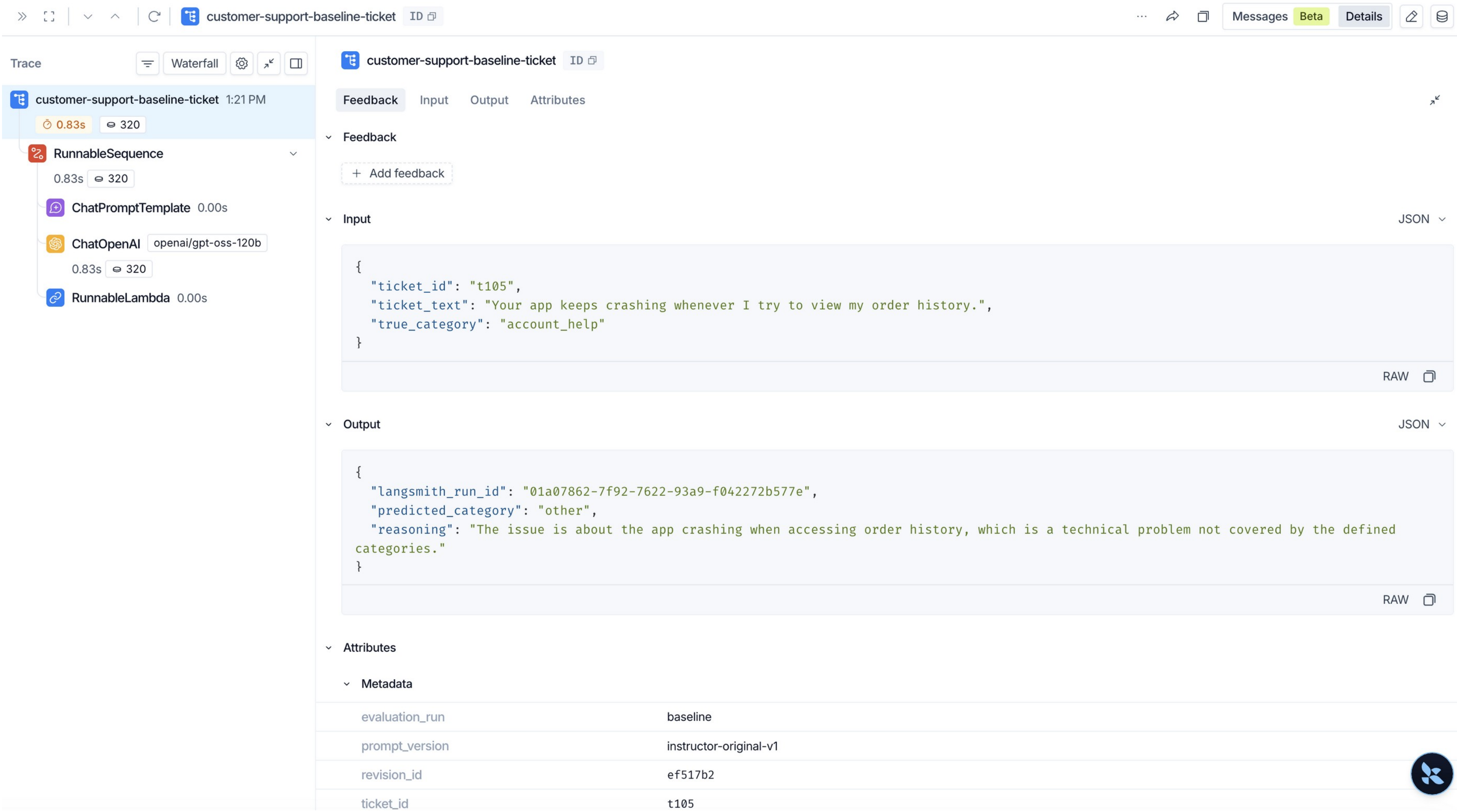

Figure 1 Baseline trace for ticket t105

LangSmith Improved Trace

The improved run used the same ticket and model. The prediction changed to account_help, and the reasoning explicitly cited the new disambiguation rule.

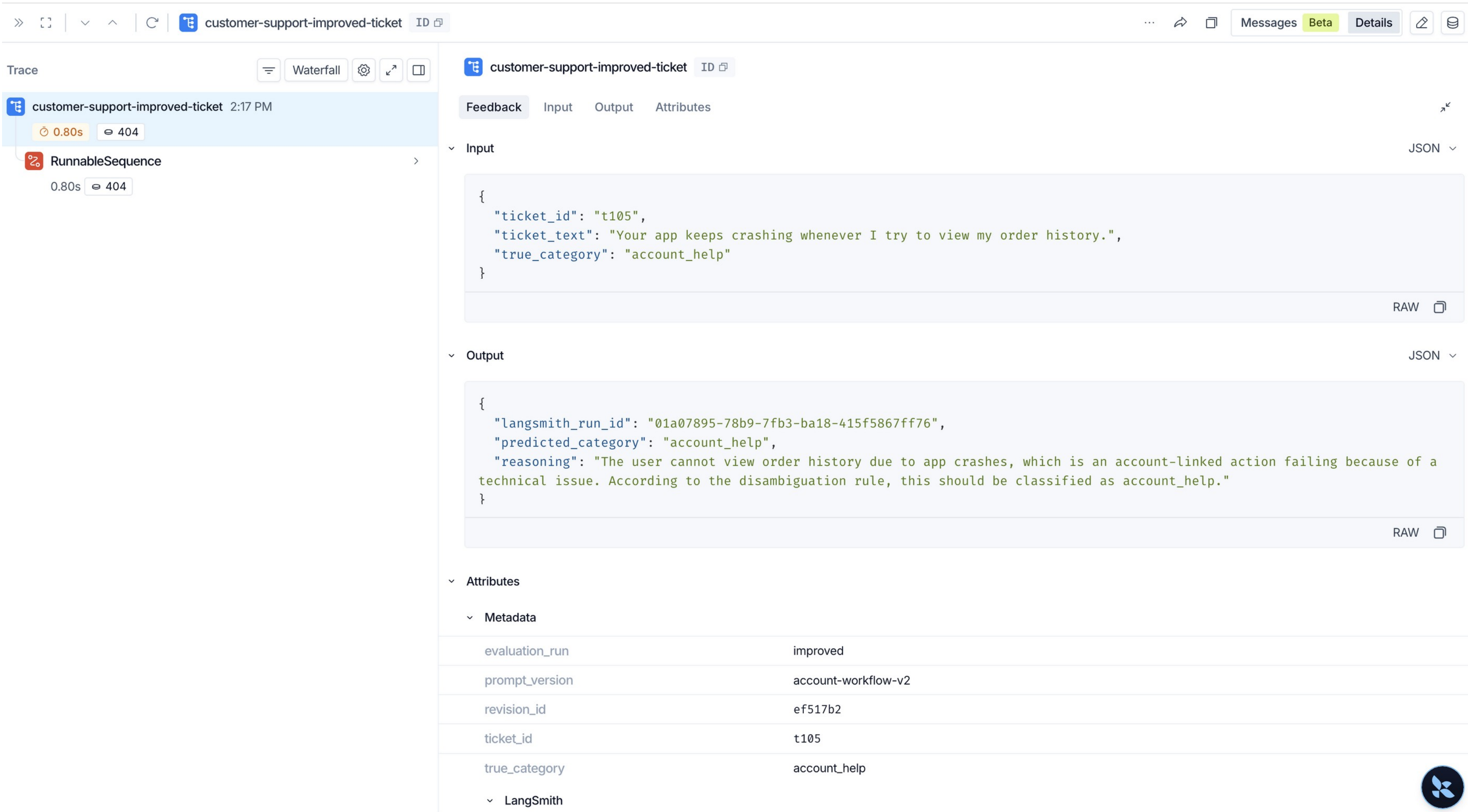

Figure 2 Improved trace for ticket t105

Trace Findings

Case	Baseline	Improved	Finding
t105 app crash	other	account_help	The new rule covered account history access
t114 promo code	other	account_help	A rejected checkout code moved into account help
t117 missing record	order_status	account_help	A missing account record no longer looked like tracking
t050 price adjustment	refund_request	other	One regression exposed a separate refund boundary gap

System Health

The inspected traces completed successfully and contained the expected prompt model and parsing spans. The incorrect decisions were policy failures rather than API execution or output parsing failures.

LLM As A Judge

Judge Task

The optional binary judge evaluated all 29 instructor reference failures. It saw the customer message and the classifier's incorrect predicted intent but did not see the human failure label. It returned TRUE when the error matched Account Workflow Missed and FALSE otherwise. The human labels contained nine TRUE and 20 FALSE cases.

Model Comparison

Judge model	Agreement	p50	p95	Tokens	Decision
openai gpt oss 120b	100% 29 of 29	0.772 s	1.119 s	13,945	Selected
Qwen3 235B	100% 29 of 29	0.927 s	1.283 s	10,046	Comparison

Both models achieved perfect precision recall and F1 for both classes. No prompt calibration was justified because there were no mismatches. We selected gpt oss 120b because it was faster and gave shorter explanations. Qwen used fewer tokens but was slower and more verbose.

Representative Judge Decisions

Case	Human	Judge	Reasoning check
t103	TRUE	TRUE	Repeated logout during payment met both rubric conditions
t024	FALSE	FALSE	A wrong delivery address triggered the delivery exclusion

Regression and Next Evaluation Cycle

Remaining Regression

Ticket t050 asks whether an item that went on sale after purchase qualifies for a price adjustment. The improved prompt changed its prediction from the correct refund_request route to other. The model reasoning states that a post purchase price adjustment does not fit the listed categories.

Reason for Deferring the Fix

Fixing t050 requires a second prompt intervention that explicitly maps post purchase price adjustments to refund_request. Adding it to the current run would combine two changes and weaken attribution of the measured improvement. We therefore documented the regression and reserved the fix for a new prompt version.

Recommended Next Experiment

Step	Action	Regression check
1	Add one price adjustment rule to refund_request	Keep the account workflow rule unchanged
2	Rerun the same 150 tickets as prompt version 3	Compare every case with version 2
3	Inspect all discount related cases and prediction flips	Separate post purchase adjustments from pre purchase discounts
4	Accept the rule only if t050 is fixed without new errors	Retain trace IDs latency and token metrics

Production Monitoring

Monitor labeled routing accuracy below 95 percent, recurrence of account-workflow failures, regression rate above 1 percent, p95 latency above 1.5 seconds, and token usage more than 25 percent above the improved baseline.

Conclusion

The project produced a measurable classifier improvement and a reusable evaluation workflow. Human review identified the error structure. The fixed dataset made the experiment comparable. LangSmith connected aggregate metrics to individual decisions. Regression testing exposed the remaining refund boundary gap. The judge comparison showed how to validate an automated evaluator against human labels before relying on it.